

Grade 6 Accelerated
Unit 9
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

For questions 1–7, complete the table below.

	Data Description	Examples <i>*Answers may vary</i>	Data Classification
1.	Number of siblings a student has	0, 1, 2, etc.	numerical
2.	Breed of cat in a pet store	Tabby, Persian, Siamese, Bengal, Sphynx	categorical
3.	<i>Favorite day of the week</i>	<i>Sunday, Monday, Tuesday, etc.</i>	categorical
4.	<i>Number of pairs of shoes you own</i>	<i>2, 4, 22, 30</i>	numerical
5.	Shoe size of basketball players	10, 10.5, 11, 11.5, 12, 13	numerical
6.	Color of shoes of basketball players	<i>Red, black, white</i>	categorical

7. Numerical data, or quantitative data, is a type of data that is measurable. It is collected in the form of numbers.

For questions 8–10, classify each as Categorical or Numerical.

8. Color of shirts people are wearing in a movie theater.

Categorical

9. Make of car each teacher at your school drives.

Categorical

10. Weight of each pet you own.

Numerical

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Lesson 2 Homework

Name Answer Key
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For questions 1–7, determine whether each question is statistical or not statistical.

1. How many cousins do you have?

Statistical

2. What country is North of the United States of America?

Not statistical

3. What time does Math class start?

Not statistical

4. What shoe size do you wear?

Statistical

5. What is your favorite food?

Statistical

6. What is the capital of Florida?

Not statistical

7. The question "What is the largest Alligator length ever recorded?" is not statistical. Use the definition to justify why it is not a statistical question.

To be statistical, it needs to be a question that results in the collection of data with variability. The answer to the above question does not vary. There's only one correct answer.

8. Decide if the question, "How tall are you?" is statistical or non-statistical. Explain your decision.

Statistical because the data collected will vary because not everyone is the same height.

9. Create an example of a statistical question.

How much time do you spend on social media apps?

10. Create a counterexample that is NOT a statistical question.

How many worldwide users does Instagram have?

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Lesson 3 Homework

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Jillian took a survey of her class to find out how many sports teams each student joined throughout the last year. Use the data in the table for questions 1–3.

Number of Sports Teams Joined									
0	1	6	2	4	3	0	2	4	2

1. Determine the mean of the data.

$$\frac{24}{10} = 2.4$$

2. Interpret the mean in terms of the context.

In Julian's class, students join an approximate average of 2 sports teams.

3. What is the median number of sports teams students joined?

2

For one week Ruth kept track of how long it took for her in minutes to walk to school. Use the data for questions 4–5.

Day of the week	Total Minutes
Monday	25
Tuesday	22
Wednesday	26
Thursday	18
Friday	20

4. Find the mean.

$$\frac{111}{5} = 22.2 \text{ minutes}$$

5. Circle the data piece that is closest to the mean amount of time she spent walking to school.

6. To calculate the mean, add all the values in the data set and then divide by the total number of data values.

Florida is one of the largest producers of citrus fruits in the United States. The table below shows the amount in millions of citrus fruits produced by season. Use the data for questions 7–8.

Season	Amount of Fruit Produced (in millions)
2001 – 2002	12.824
2002 – 2003	11.206
2003 – 2004	13.045
2004 – 2005	7.597
2005 – 2006	7.832
2006 – 2007	7.236
2007 – 2008	9.119

7. Find the mean of the data.

$$\frac{68.859}{7} = 9.837$$

8. Interpret the mean amount of fruit produced based on the data.

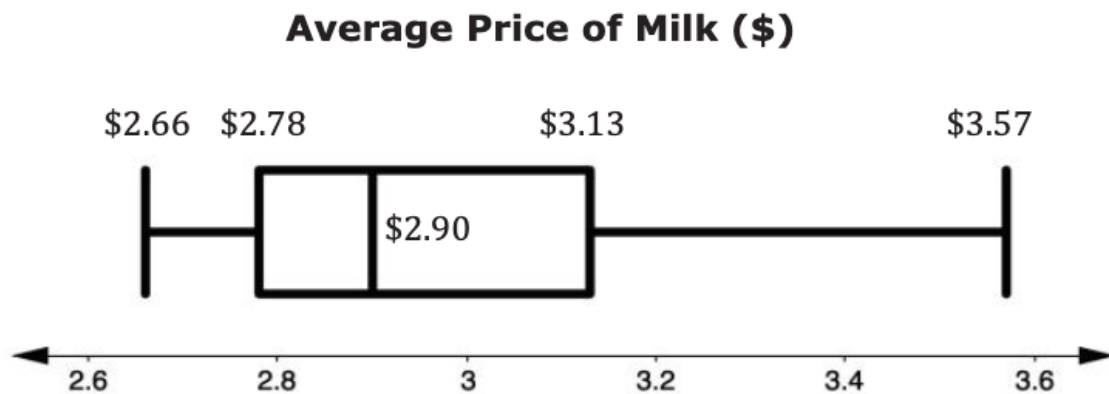
The average yearly amount of citrus fruits was 9.837 million.

9. The Tampa Bay Rays had 6 outfielders in their 2021 season. Find the mean of their ages.

Pitchers Name	Pitchers Age
Randy Arozarena	27
Kevin Kiermaier	31
Josh Lowe	24
Manuel Margot	27
Austin Meadows	26
Brett Phillips	27

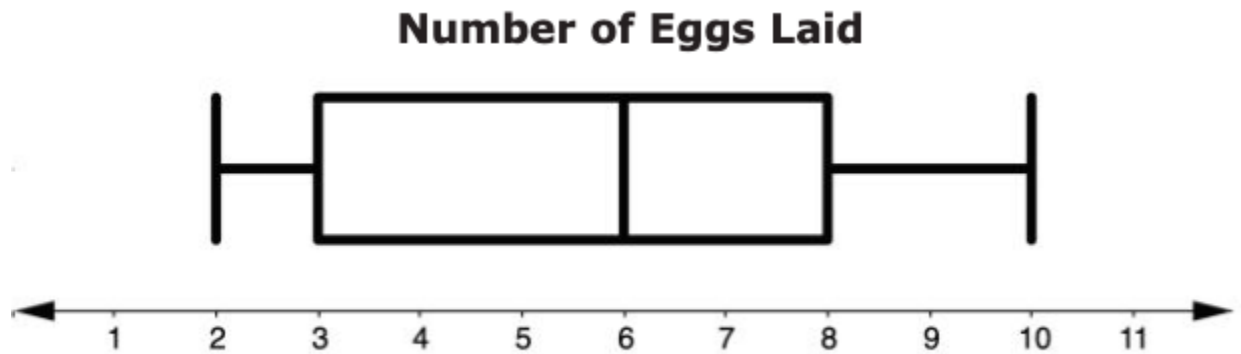
10. Circle the data pieces that are closest to the mean.

The average price of milk, in dollars (\$), for each month for 7 years was recorded and used to create the box plot shown. Use the box plot for questions 1-6.



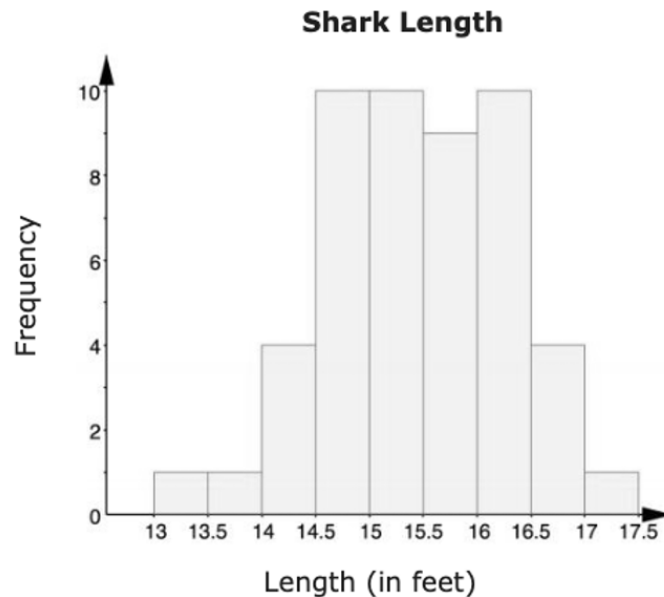
1. The minimum price is \$2.66, while the maximum price is \$3.57.
2. The median price is \$2.90.
3. The upper quartile is \$3.13.
4. The lower quartile is \$2.78.
5. The range is \$0.91.
6. The interquartile range is \$0.35. The IQR was calculated by subtracting Q_1 from Q_3 .

A farmer recorded the number of eggs her farm full of hens laid in one week. The box plot represents the number of eggs laid by each hen during that week. Use the box plot for questions 7-10.



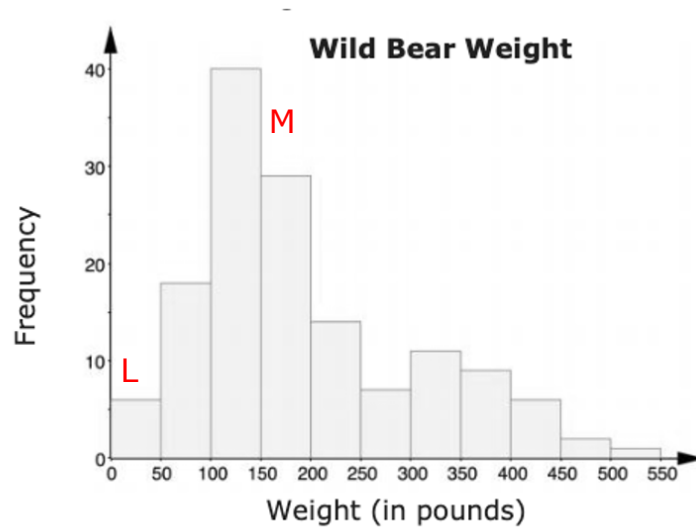
7. Write a description of the spread and distribution of the number of eggs laid.
The least number of eggs laid was 2, while the most eggs laid was 10. The range was 8, which is a large spread for egg laying. The median amount of eggs laid was 6.
8. What percentage of hens laid 2 or more eggs?
100%
9. What percentage of hens laid between 6 and 10 eggs?
50%
10. What percentage of hens laid between 2 and 8 eggs?
75%

The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.



- How many sharks are between 16.5 and 17 feet?
4
- How many sharks measured are less than 15 feet?
16
- What is the smallest possible length of a shark in this data set?
13 ft
- Are there more female sharks measuring less than 15.5 feet or more than 15.5 feet?
More female sharks measuring less than 15.5 ft.
- If sharks younger than adult age were added to the sample, hypothesize what would happen to the spread of the data?
It would become skewed because there'd be more data at 13 feet and under.

In a study of wild bears, researchers measured the weights, in pounds (lb.) of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



6. Is it possible that the sample contains a bear that weighs 50 pounds?
Yes
7. More than half of the bears weigh less than 200 pounds in this study.
8. Mark an "M" above the bar that contains the approximate median.
9. Mark an "L" above the bar that contains the bear that weighs the least.
10. Which single range of weights has the most bears?
100 – 150

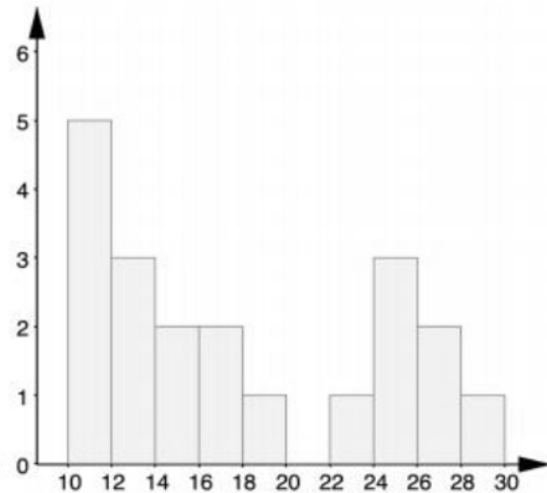
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Lesson 6 Homework

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1. A left skewed graph has a longer trail of data to the left and the peak of the data on the right.

2. The shape of the histogram is

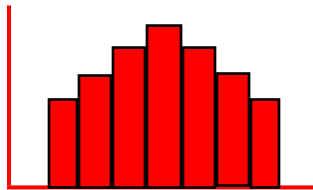
- ☐ symmetric.
- ☐ left skewed.
- ☒ right skewed.



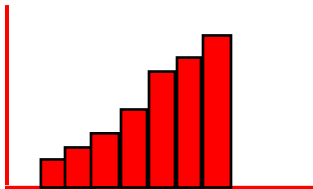
3. If the data value 40 were added to the histogram above, it would be considered an outlier.

Answers may vary

4. Sketch a histogram whose shape can be described as symmetric.



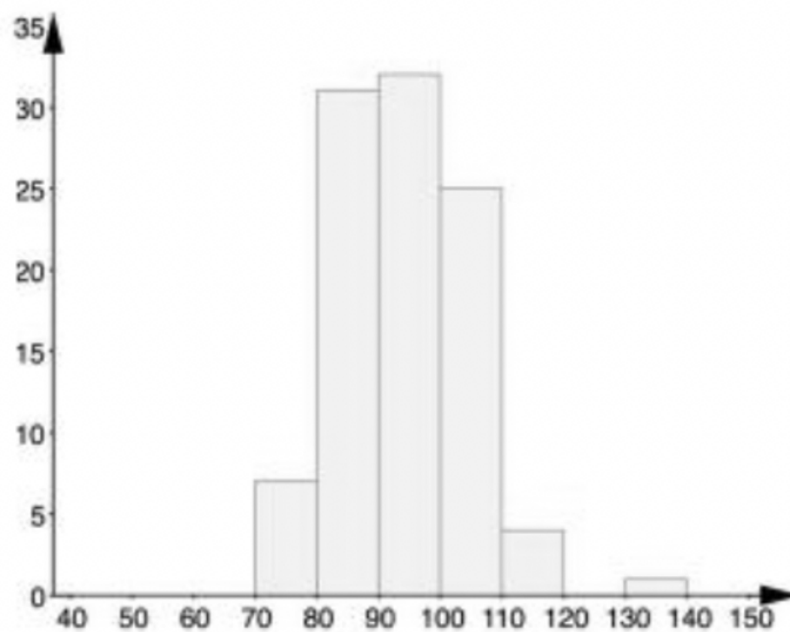
5. Sketch a histogram whose shape can be described as left skewed.



6. If an outlier is added to a data set that is far less than the median, which direction will the graph be skewed?

Left skewed

Use the histogram for questions 7–10.



7. The shape of the histogram can be described as

- ☐ symmetric.
- ☐ left skewed.
- ☒ right skewed.

8. Is there a gap in the histogram? If so, where?

Yes, from 120 – 130

9. Is there a cluster in the histogram? If so, where?

Yes, from 70 – 120

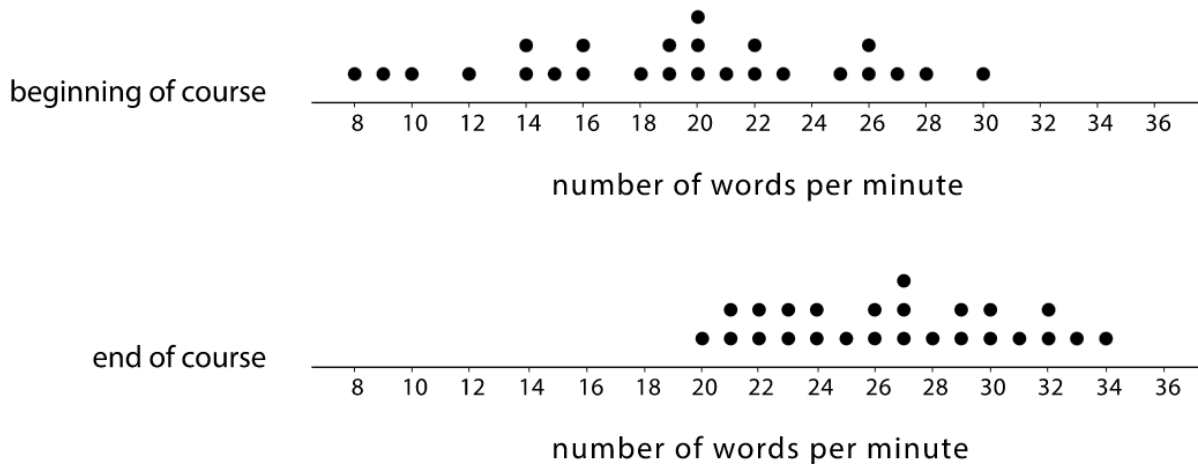
10. If the data value 40 were added to the histogram above, it would be considered an outlier.

Answers may vary

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Lesson 7 Homework

Name **Answer Key** _____
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The teacher recorded the number of words that her students could type per minute at the beginning of a course and again at the end. The two dot plots show the two data sets.



1. How many students were in the course?

25

2. Determine the mean number of words typed per minute at the beginning of the course.

$$\frac{480}{25} = 19.2$$

3. Determine the mean number of words typed per minute at the end of the course.

$$\frac{666}{25} = 26.64$$

4. What percentage of students typed 22 words per minute or more at the beginning of the course versus the end of the course?

Beginning

End

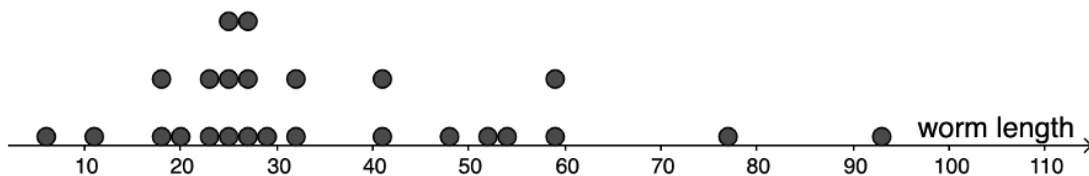
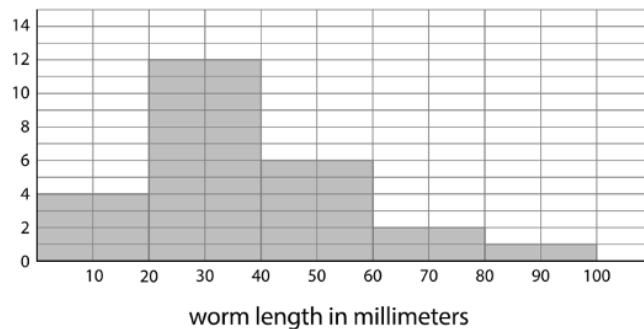
$$\frac{9}{25} = 36\%$$

$$\frac{22}{25} = 88\%$$

5. Describe the spread and variability of the data from the beginning of the course to the end of the course.

At the beginning, the range was 22, and at the end it was 14. The average words typed per minute increased by 7 wpm by the end with a greater number of students being able to type 22 wpm or more by the end.

An earthworm farmer set up several containers of a certain species of earthworms so that he could learn about their lengths. The lengths of the earthworms provide information about their ages. The dot plot and histogram display the measure of the lengths of earthworms in one of the containers. Each length was measured in millimeters.



6. To determine the range, which data display should be used and why?

The dot plot because you can actually see lengths and not just ranges

7. What do you notice about the shape of the distribution?

The shape is slightly skewed right

8. How many earthworms were measured?

25

9. Which data display can be said to determine how many earthworms measured 25 millimeters?

Dot plot

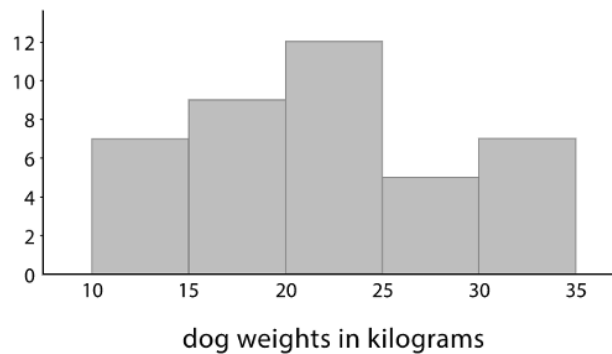
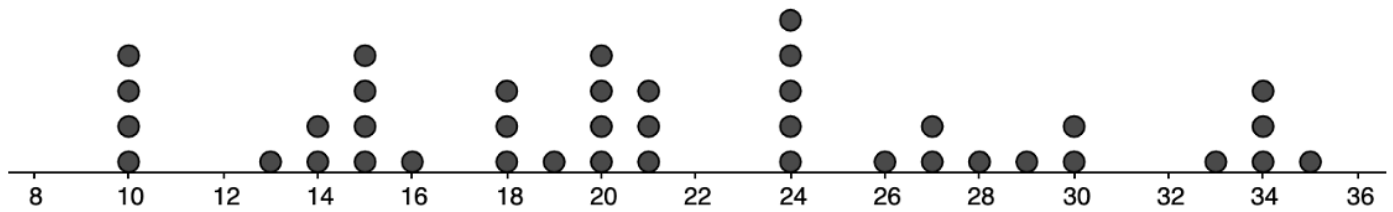
10. How many earthworms measured 25 millimeters?

3

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Lesson 8 Homework

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A histogram and dot plot show the weights, in pounds, of 40 dogs at a dog show.



1. Which data display can be used to determine the mean weight?

Dot plot because you can see actual data

2. Find the mean weight. Interpret the mean.

$$\frac{860}{40} = 21.5$$

The average weight of the dogs was 21.5 lbs.

3. Which data display can be used to determine the median weight?

Dot plot

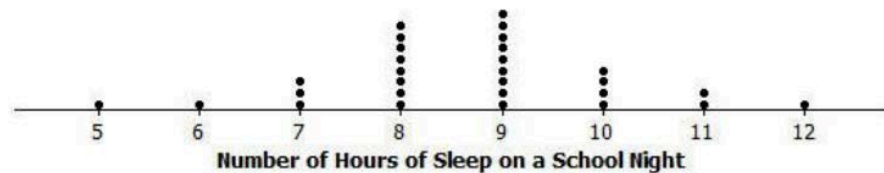
4. Find the median weight. Interpret the median.

$$\frac{20+21}{2} = 20.5$$

The middle value of the dogs' weights was 20.5 lbs. Half of the dogs weighed more than 20.5 lbs., and half weighed less than 20.5 lbs.

Rocky is a sixth grader at Roosevelt Middle School and investigated the number of hours of sleep sixth-grade students get on school nights. He surveyed 29 sixth graders and organized his data on the dot plot.

Dot Plot of Number of Hours of Sleep



5. Describe the spread and shape of the distribution.

Symmetrical with a range of 7 hours

6. Find the mean number of hours slept by 6th graders.

$$\frac{251}{29} = 8.7 \text{ hours}$$

7. Interpret the mean.

The average amount of hours a sixth grade student sleeps is 8.7 hours.

8. Find the median.

9

9. Interpret the median.

The middle value of the dataset is 9 hours.

10. Suppose Rocky surveyed 10 more students and they reported the following (in hours); 9, 10, 9, 9, 12, 12, 12, 8, 10, 7. If these 10 data points were added to the dot plot, explain how would the mean and median change?

$$\text{Mean: } \frac{34.9}{39} \approx 8.9 \quad \text{Median still} = 9$$

With more data, the mean and median come closer to being the same value.

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The ages of 28 people in a movie theater are shown.

6, 6, 7, 7, 7, 8, 8, 8, 8, 9, 9, 9, 10, 10, 11, 13, 14, 18, 19, 25, 27, 28, 28, 33, 37, 40, 45, 67

1. Determine the five-number summary for the data set.

Minimum: 6

First Quartile: 8

Median: 10.5 $\left(\frac{10+11}{2}\right)$

Third Quartile: 27.5 $\left(\frac{27+28}{2}\right)$

Maximum: 67

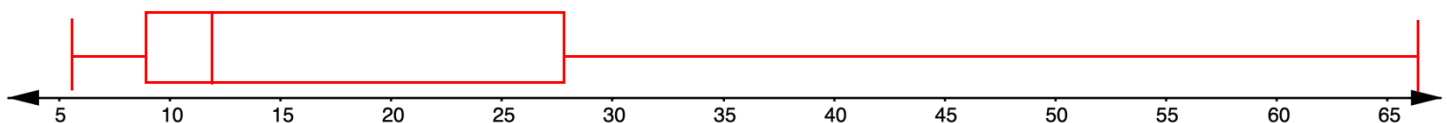
2. What is the range of the data?

$$67 - 6 = 61$$

3. What is the interquartile range of the data?

$$27.5 - 8 = 19.5$$

4. Create a box plot of the ages at the movie theater.



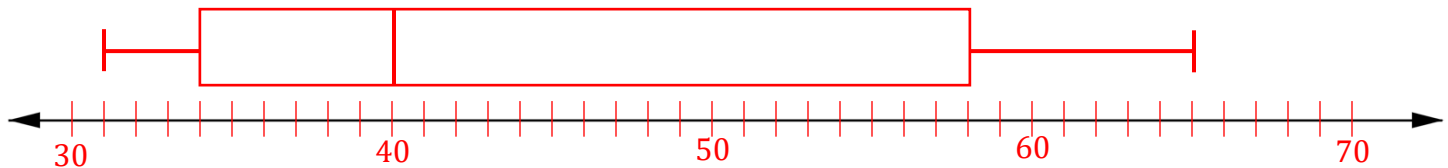
A bakery's data on the number of muffins sold each day for the last 14 days are shown.

32	41	65	58	31	38	46	39	38	62	60	34	31	44
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5. Determine the five-number summary for the data set.

Minimum: 31 Q_3 : 58
 Q_1 : 34 Maximum: 65
 Median: 40

6. Create a box plot of the bakery's daily muffin sales.



7. Explain how you determine the scale to use for the number line of your box plot.
 I counted by 1's because there was enough room and because the range was only 34.

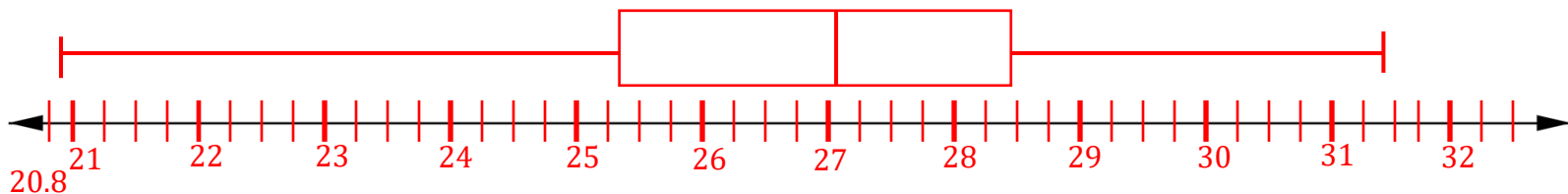
LeBron James's average points per game (PPG) for each regular season that he's been in the NBA are shown.

Season	03-04	04-05	05-06	06-07	07-08	08-09	09-10	10-11	11-12	12-13
PPG	20.9	27.2	31.4	27.3	30.0	28.4	29.7	26.7	27.1	26.8
Season	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	
PPG	27.1	25.3	25.3	26.4	27.5	27.4	25.3	25.0	29.0	

8. Determine the five-number summary for the data set.

Minimum: 20.9 Q_3 : 28.4
 Q_1 : 25.3 Maximum: 31.4
 Median: 27.1

9. Create a box plot of LeBron James's average points per game.



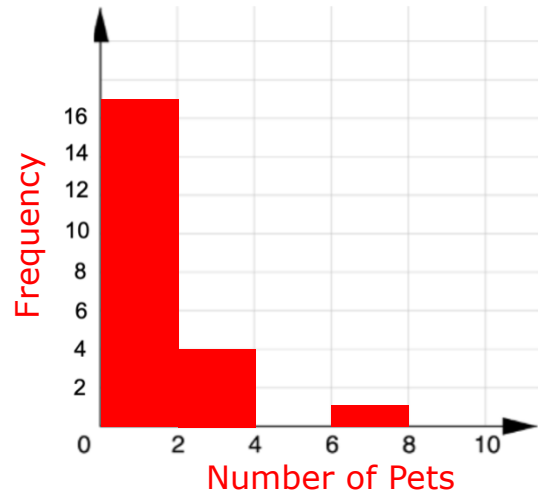
10. What is the interquartile range?
 $28.4 - 25.3 = 3.1$

Luna collected data on the number of pets her classmates have. Her data is shown.

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 3, 3, 7

1. Create a frequency chart of the data.
2. Create a histogram of the data.

Number of Pets	Frequency
0 – 2	17
2 – 4	4
4 – 6	0
6 – 8	1

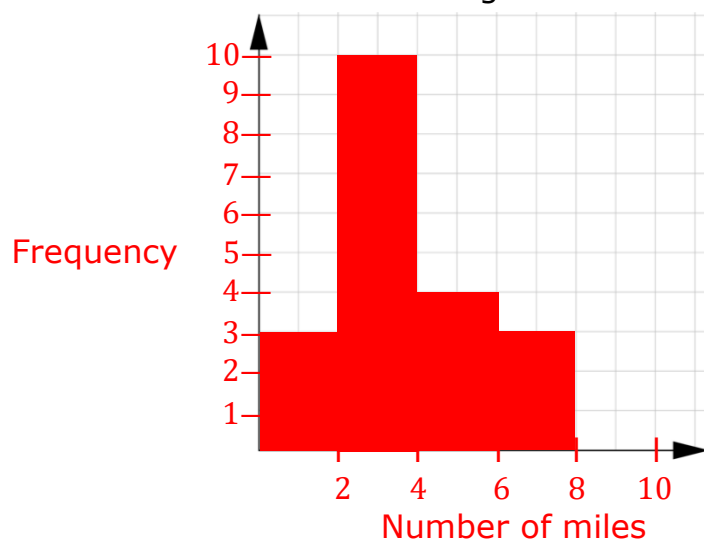


Marcus is training for a half-marathon. His daily miles run for the last 20 days are shown.

3.5	3	2.5	3.5	6	0	3.5	4	2.5	2
3	6.5	0	4	4	3.5	4.5	2	7.5	0

3. Create a frequency chart of the data.
4. Create a histogram of the data.

Number of Miles	Frequency
0 – 2	3
2 – 4	10
4 – 6	4
6 – 8	3



5. Explain how you determined the scale for the axes on your histogram in question 4. **Answers may vary. The scale for the x-axis is 2 to match the interval width of 2: 8 (estimate of highest data value) $\div$ 4 (rows in the frequency chart) $= 2$. No frequency is > 10 , so I could number the y-axis by 1's.**

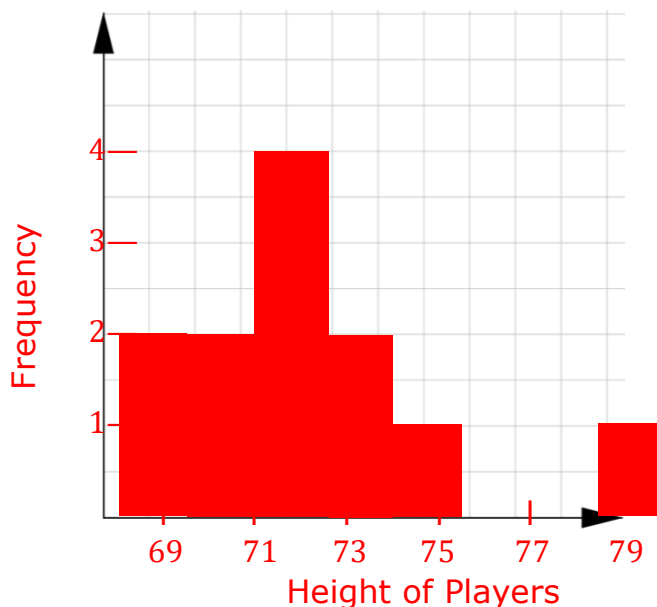
The heights, in inches, of 12 players on a college basketball team are shown.

71	73	70	79	68	70	73	72	68	74	72	71
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6. Create a frequency chart of the data.

Height (in.)	Frequency
68 – 69.5	2
69.5 – 71	2
71 – 72.5	4
72.5 – 74	2
74 – 75.5	1
75.5 – 77	0
77 – 78.5	0
78.5 – 80	1

7. Create a histogram of the data.



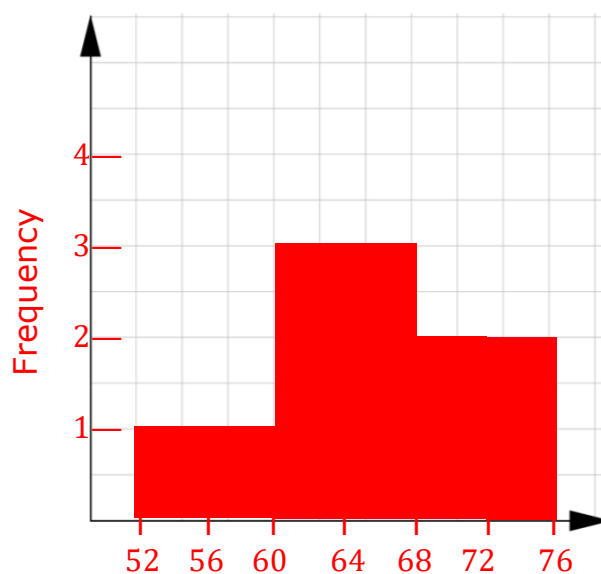
The heights, in inches, of 12 players on a sixth-grade basketball team are shown.

64	73	60	67	68	58	72	53	62	65	70	61
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8. Create a frequency chart of the data.

Height (in.)	Frequency
52 – 56	1
56 – 60	1
60 – 64	3
64 – 68	3
68 – 72	2
72 – 76	2

9. Create a histogram of the data.



10. How are your histograms for questions 7 and 9 different?

Answers may vary.

7 has one peak and an outlier and 9 is more symmetrical without one peak.